STUDENT GUIDE · READ ALONGSIDE THE ANIMATION

How to read the IS256 × ica locus animation

One film, one cell, one decision made twice. The insertion sequence IS256 jumps into the *icaC* gene of *S. aureus*, switching biofilm formation off; it then jumps out again, restoring it exactly. The whole film is a 36-second loop in four nine-second chapters — ON, JUMP IN, OFF, JUMP OUT. Timecodes like `0:17` are the animation's own clock: drag the timeline scrubber to them, or click a chapter dot.

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1 · The colour key

THE ONE RULE THAT DECODES EVERYTHING

Two colour stories run in parallel. Up top, colour tells you *whose DNA it is and where it sits*: blue arrows are the host's *ica* genes, and the red block is IS256 — watch which site it occupies. Down in the cell, colour tells you *the chemical state of the polymer*: dark-navy beads are acetylated GlcNAc inside the cell; they turn green at exactly one moment — when a unit passes **IcaB on the way out**. When a bead changes colour, chemistry has happened: that is deacetylation, and it is what makes surface PNAG different from the chain being built inside.

DNA and the element

	WHAT IT IS	WHERE YOU SEE IT
	ica operon genes	The blue arrows along the top: *icaA, icaD, icaB, icaC*, one co-transcribed unit. While IS256 is in situ, the *icaC* arrow splits into two faded fragments flanking it — the region gets longer than the intact gene.
	IcaR repressor	The violet arrow and small dimer on its operator, between *icaR* and *icaA*. It keeps the operon off until released; this film runs with the operon already ON.
	IS256 element	The red block with dark-red end caps, ~1.3 kb in reality. Starts at its home site (right), flies to the *icaC* hot spot, then back.

TSD — target-site duplication	The small cyan dots. Two appear at the two junctions of inserted IS256 (8 bp duplicated once per end); one marks the hot spot on the home site while the element is away.
Surface	The grey bar at the bottom — a medical-device surface. Biofilm cells sit on it; when PNAG disappears they drift off.

The polymer and its machinery

	WHAT IT IS	WHAT IT DOES, AND WHEN
	PNAG (GlcNAc, inside)	The dark-navy beads linked by a backbone line — acetylated monomers of the chain being built. Inside the cell only.
	Deacetylated PNAG (coat)	The green beads — the same units after IcaB strips their acetyl groups on exit. Also the coat rings around the cell and the soft halos on biofilm cells.
	IcaA·D — transferase	The teal membrane block where the chain is built, fed by UDP-GlcNAc. A peristaltic beat every couple of seconds pushes one new unit in at the growing end and advances the whole strand.
	IcaC — the channel	The violet box straddling the peptidoglycan wall. The only door out of the cell for this polymer; a red × appears over it while IS256 sits in its gene.
	IcaB — deacetylase	The amber block sitting *outside* the cell. It works as each unit crosses out past it, and flashes a halo when it does. Navy→green happens here, not at the membrane.
	UDP-GlcNAc	The lavender pill — the substrate supply feeding into IcaA·D. Present in every chapter.
	icaADBC mRNA	The amber dashed line flowing from the operon to the cell — and it keeps flowing in every chapter, even while *icaC* is broken. It never gets a × (no polar effect).
	Blocked function	The red ×. Over the downstream *icaC* fragment and over the IcaC channel while IS256 is in situ; beside the PIA/PNAG chip when the coat is gone.

TWO THINGS WORTH TELLING APART

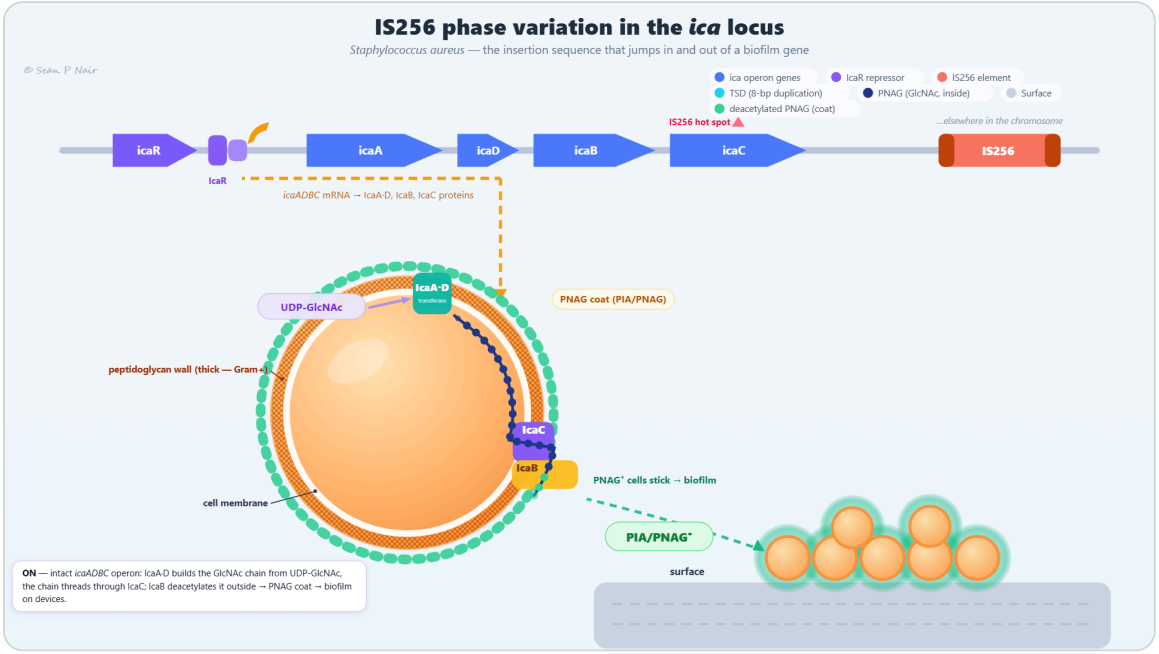
***icaC* (lowercase, top of frame) and IcaC (capitalised, in the cell wall) are not the same object.** The gene is DNA; the protein is the channel it encodes. Inserting IS256 breaks the *gene*, but a working channel needs a functional transcript to be made — and on excision the reverse lag shows: the gene reads as one closed arrow the instant IS256 takes off, yet the ✕ stays over the IcaC channel until the element lands back home. Genetics can change faster than the protein machinery does.

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2 · Chapter by chapter

S. aureus, one cell and its neighbourhood on a device surface. Four chapters, nine seconds each; the loop is seamless — frame 0 is identical to frame 36.

01 · ON 0:00



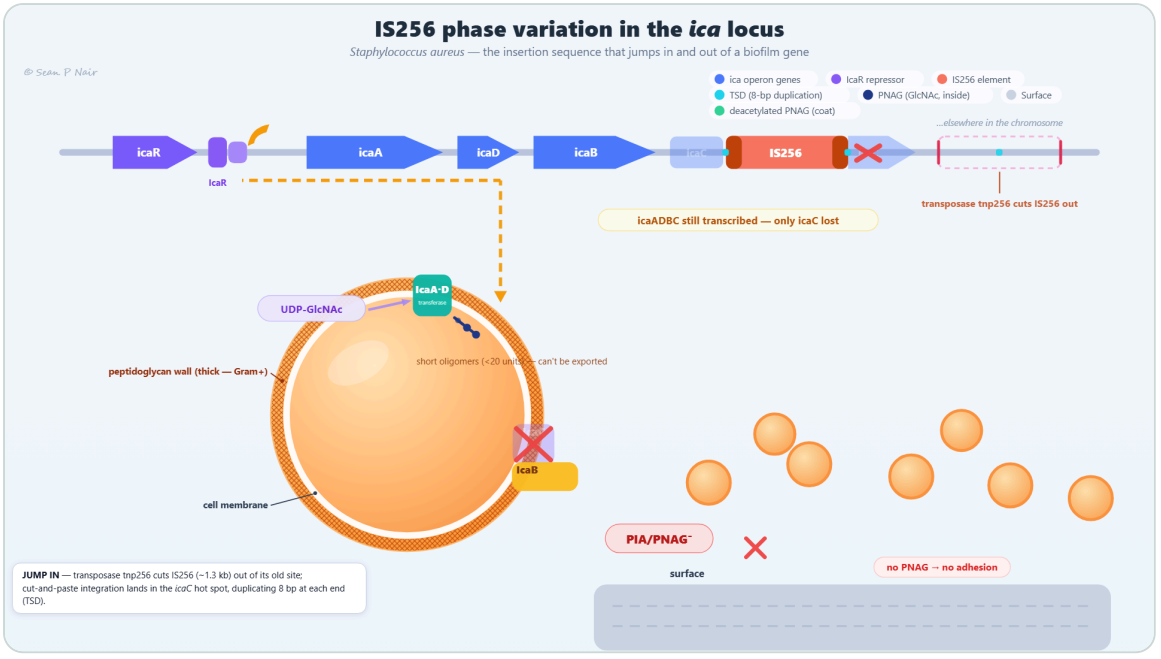
0:05 — steady state. The chain runs from IcaA·D (teal) down through the IcaC channel (violet) past IcaB (amber, outside); navy inside, green from IcaB's exit on. The coat rings are layered, and PNAG⁺ cells with halos are stuck to the device surface.

The intact *icaADBC* operon is transcribed — the amber dashed mRNA line runs from the gene arrows into the cell throughout. The IcaA·D transferase builds a chain of β-(1→6)-linked GlcNAc units from UDP-GlcNAc, adding one at the growing end with each peristaltic beat; the whole strand advances in lockstep so every bead travels the same path: anchor → channel → exit.

Units stay navy (acetylated) while inside and turn green (deacetylated) exactly as they cross out past IcaB — a halo flash marks each conversion. Exported units join the PNAG coat, which thickens layer by layer during the chapter; coated cells adhere to the device surface and stack into a biofilm.

Look for: the navy→green switch happens at IcaB, not at the membrane — and in steady state exactly three trailing beads are green. The coat rings advance one bead-pitch per beat, in step with the secreted chain.

02 · JUMP IN 0:09



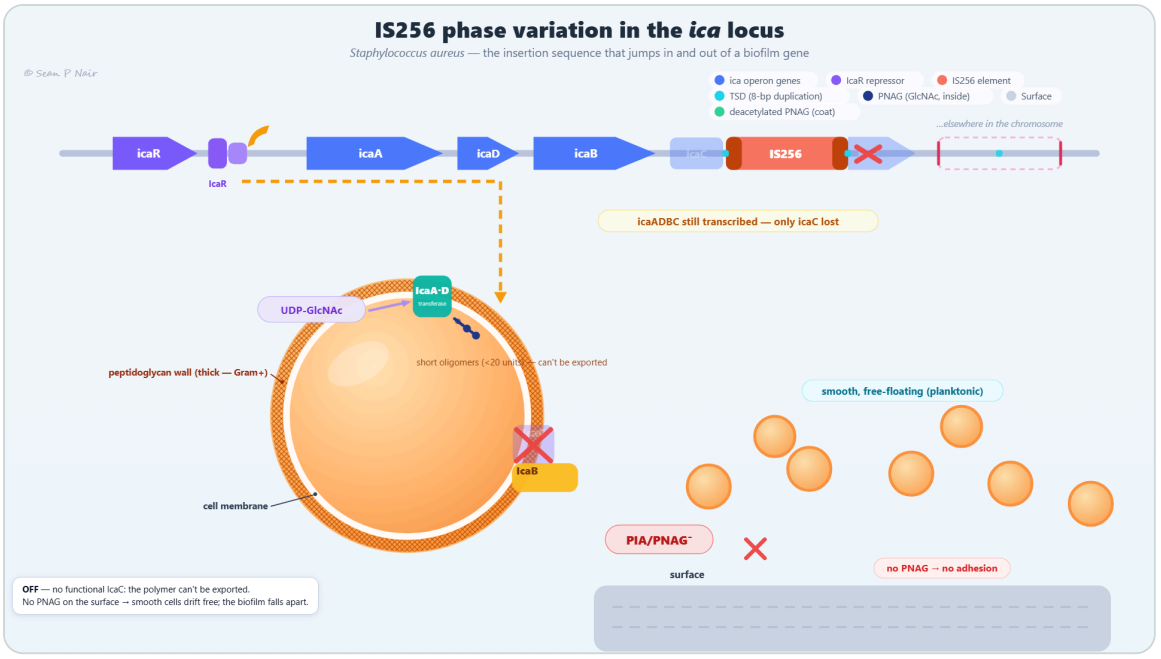
0:17 — IS256 (red) is seated in the *icaC* hot spot, which now reads as two faded fragments flanking it. The small cyan TSD dots mark the 8 bp duplicated at each junction; the home site on the right remains as a dashed ghost with nicks. Inside the cell only a short stub of chain is left — "short oligomers (<20 units) — can't be exported".

The transposase `tnp256` cuts IS256 out of its home site and carries it, cut-and-paste, to the documented hot spot inside *icaC*. Insertion duplicates 8 bp of target DNA once at each end — so two identical TSD copies flank the element (the two cyan dots), and the gene is not replaced but *split*: the icaC–IS256–icaC region is longer than the original gene.

Here is the counterintuitive part. Because *icaC* is the **last** gene of the operon, breaking it has no polar effect: transcription keeps running (the amber pill says so — "icaADBC still transcribed — only icaC lost"), IcaA·D and IcaB stay up, and chain synthesis continues. But with no functional IcaC channel there is no door out of the cell, so the chain retracts to a short trapped oligomer. The coat fades, cells lose adhesion and drift free.

The trap is at export, not synthesis. If you saw "operon off" written on this chapter it would be wrong: the arrows are lit, the mRNA line flows, substrate keeps feeding in. The polymer is still being made — it simply cannot leave. That distinction (gene intact vs protein functional) is what makes the JUMP OUT chapter's lag meaningful.

03 · OFF 0:18

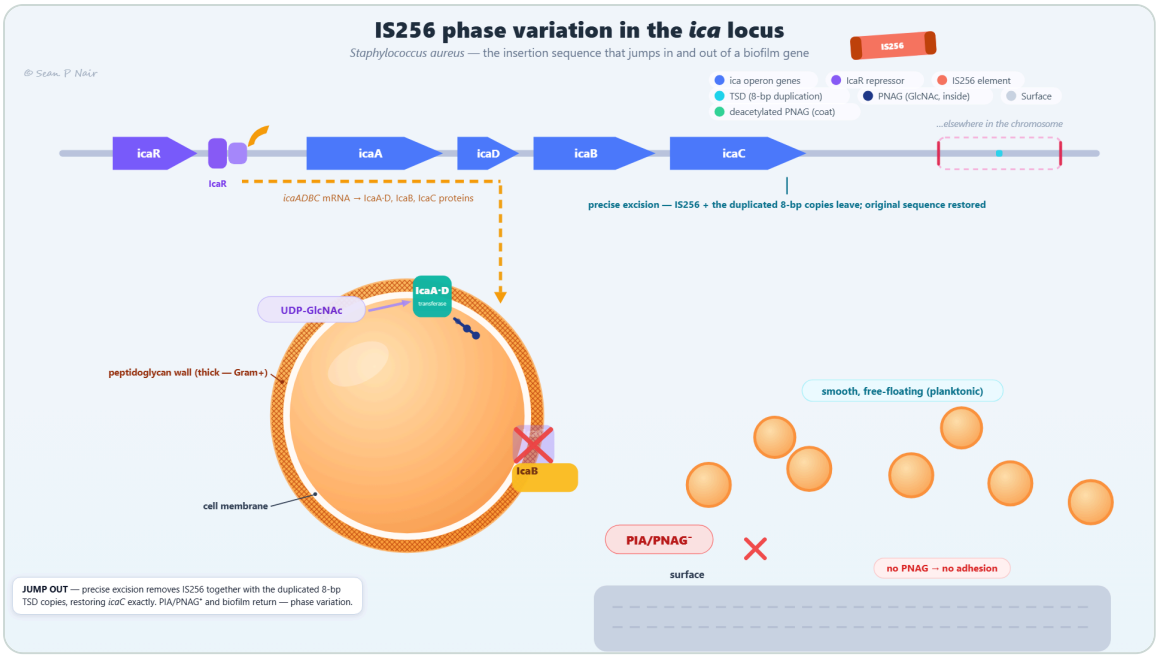


0:24 — the settled OFF phenotype. No coat rings, no halos: the cells are smooth and free-floating (planktonic) above a bare surface. The PIA/PNAG⁻ chip carries a red ✕ — "no PNAG → no adhesion".

This is the phase-variant phenotype: smooth, non-adherent, free-drifting cells. In a real culture these are not a separate species or a new mutation lineage — they are the same clone with IS256 in *icaC*, and they can revert at any division.

Do not read "OFF" as "the operon is off". The blue gene arrows stay lit, the amber mRNA line keeps flowing, UDP-GlcNAc still feeds IcaA·D. Everything upstream of export works; only the IcaC channel is missing. If a question asks where synthesis is blocked in this chapter, the answer is *the membrane exit*, not transcription or polymerisation.

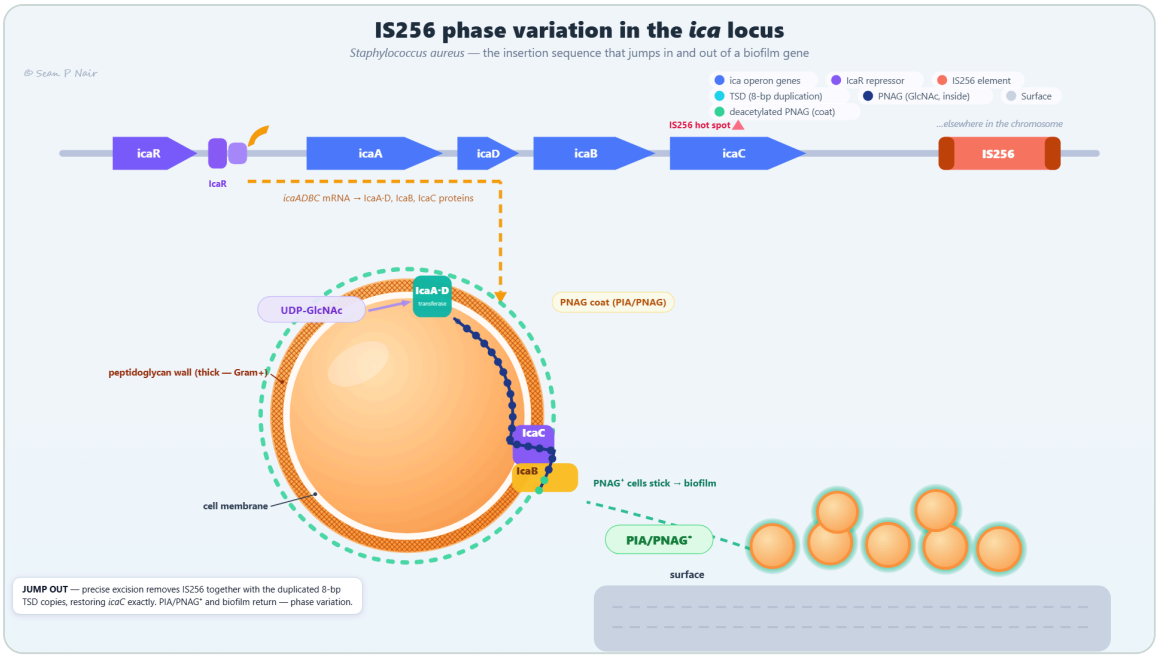
04 · JUMP OUT 0:27



0:29 — IS256 in flight home. The *icaC* arrow has closed into a single gene again the instant the element took off ("precise excision — IS256 + the duplicated 8-bp copies leave; original sequence restored"), but the ✕ still sits over the IcaC channel below.

Precise excision cuts out IS256 *together with one copy of each duplicated 8-bp pair*, so the original single copy remains and *icaC* is restored base-for-base — no scar, no leftover. The cyan TSD dots disappear from the gene as the element departs: they leave with it.

Note the lag: the gene reads intact immediately at take-off, but the channel ✕ persists until IS256 lands home — the broken protein machinery turns over more slowly than the genome changes. Once the channel reopens, the chain re-threads anchor-first through IcaC, the coat rings rebuild layer by layer, and the cells return to the surface one by one as their halos bloom.



0:34 — restored and regrown. IS256 is home (hot-spot mark over *icaC*), the full chain threads IcaC again, coat rings are rebuilding, and the biofilm is reassembling on the device. A few seconds later this frame is identical to 0:00 — the loop closes.

This is phase variation in a real infection. In *S. aureus*, spontaneous IS256 insertion into *icaC* yields PIA/PNAG-negative variants that revert by precise excision at roughly $1–5\times10^{-9}$ per cell division (Kiem et al. 2004, strain SA204); in *S. epidermidis*, ~30% of PIA-negative isolates carry IS256 in *icaA/icaC* (Ziebuhr et al. 1999). One clone, two phenotypes, switching both ways — a population hedge: while the $PNAG^{+}$ majority builds biofilm on devices, the rare $PNAG^{-}$ minority stays free-floating, ready to switch back when conditions change.

3 · Where the animation simplifies

Most teaching animations are simplified somewhere. These are the places this one is, so the exaggerations are not mistaken for scale:

- **The element is drawn huge.** IS256 is ~1.3 kb inside a ~2.8 Mb chromosome; at true scale you could not see it or its 8-bp duplications. The cyan TSD dots are markers, not to-scale DNA.
- **One cell's fate is shown as the whole story.** Real phase variation is stochastic and population-level: each division carries ~10^{-9} odds of insertion or excision, so $PNAG^{+}$ and $PNAG^{-}$ cells coexist in one culture. The film follows a single representative cell through both directions.
- **The chain is a deterministic conveyor.** A constant-length rope advancing one unit per beat stands in for stochastic polymerisation and transport. What it keeps exact is the teaching point: deacetylation happens at IcaB, on exit — never inside the cell or at the membrane.
- **Seven biofilm cells** are drawn for legibility; a real film holds thousands of cells in a hydrated PNAG matrix. The soft halos stand in for that corona — there is no solid "matrix blob".
- **Time is compressed and even by design.** Each chapter gets nine seconds regardless of how long the state lasts in reality; an OFF cell can stay OFF for generations, but the film moves on so you see the round trip.

4 · Check yourself

Answers are hidden — have a go first. If you can answer these, you have followed the animation.

? IS256 sits inside icaC and the operon is still transcribed. Why has biofilm formation stopped anyway?

Because *icaC* is the last gene of the co-transcribed operon, so breaking it has no polar effect on *icaA/icaD/icaB* — those proteins are still made. But IcaC is the channel that externalises the polymer: with no functional channel the chain cannot leave the cell and retracts to a short trapped oligomer. No coat, no adhesion, no biofilm. The block is at export, not synthesis.

? What are the two small cyan dots flanking inserted IS256, and why are there exactly two?

They mark target-site duplications (TSDs). Cut-and-paste transposition cuts between two imperfectly repeated 8-bp sequences at each junction, so the target sequence is duplicated once per end — two identical copies flank the inserted element. Watch them both disappear from *icaC* when IS256 excises: they leave with it.

? Why does precise excision restore icaC exactly, instead of leaving a scar or losing a few bases?

Because insertion duplicated the target 8 bp once at each end, the in-situ region holds two copies of that sequence. Precise excision removes IS256 together with one copy of each duplication, leaving the original single copy in place — so the restored gene is base-for-base identical to the pre-insertion sequence.

? Navy beads and green beads are the same polymer. What changed, where exactly did it change, and why does that matter?

The acetyl groups: navy is acetylated GlcNAc (PIA-like), green is deacetylated PNAG. The change happens at IcaB — an enzyme sitting *outside* the cell, acting as each unit crosses out past the IcaC channel mouth. It matters because surface chemistry differs from intracellular chemistry: the same chain is only "PNAG" once it has been exported and deacetylated.

? In JUMP OUT, which comes back first — the gene or the channel? What does that tell you?

The gene: *icaC* reads as one closed arrow the instant IS256 takes off, while the ✕ over the IcaC channel lags until the element lands home. The genome changes instantly; the protein machinery has to turn over first. Phenotype can lag genotype — a general lesson in how cells work.

? A single clone of S. aureus grows both biofilm-forming adherent cells and smooth free-floating cells. Explain that without invoking two different mutations.

Phase variation: IS256 spontaneously inserts into *icaC* ($PNAG^-$, smooth) and precisely excises out of it again ($PNAG^+$ restored), at ~10^{-9} per cell division in each direction. One genotype, two reversible phenotypes — the population carries both at once as a hedge, so whichever conditions favour is already represented.

? You need to show that deacetylation happens outside the cell, not inside. What in the film proves it?

IcaB is drawn on the outer face of the membrane, and the colour split follows IcaB's exit, not the membrane: beads stay navy across the whole peptidoglycan wall and channel, then turn green — with a halo flash at each conversion — only once they have crossed out past IcaB. Even in steady state the last three trailing units are green while everything upstream is navy.

? The film shows a purple repressor (IcaR) sitting upstream, between icaR and icaA. How does one small protein switch the whole locus on and off?

All five genes are read as *one* mRNA: *icaADBC* is a single co-transcribed operon driven by one promoter in that intergenic region, where IcaR binds its 28-bp operator. So when the repressor sits on the DNA no transcript is made and none of the machinery — IcaA·D, IcaB or IcaC — exists; when it leaves, the whole line comes back together at once. Notice this is a different lever from IS256: IcaR controls whether the genes are *read* (transcriptional control), while IS256 breaks one gene's product structurally.

? Every bead on the rope is the same sugar, copied over and over. Which sugar — and what hands it to the transferase?

Glucosamine (GlcNAc), linked β-(1→6). The IcaA·D box at the top of the cell adds one unit per beat from *UDP-GlcNAc*, the pill feeding into it — a nucleotide-sugar donor. And the transferase needs both partners: IcaA on its own transfers poorly, so IcaD is its required accessory subunit (which is why the film always draws them as one box).

? In OFF, short chains still form inside the cell but never get long. What can the transferase do on its own, and what needs the channel?

IcaA·D by itself builds only *short* oligomers — under ~20 units (the film's trapped three-bead stub is that limit, drawn small for legibility). Extending them into full-length polymer and pushing it out across the membrane are both jobs of the IcaC channel. So in OFF synthesis keeps happening but nothing leaves: production without export.

? The film marks an "IS256 hot spot" over icaC. How sure can we be about why IS256 keeps landing there?

We're very sure it *does* land there — in S. aureus SA204 the PIA-negative variant carries IS256 inside *icaC*, and in S. epidermidis roughly a third of PIA-negative isolates have IS256 in the ica locus (Kiem 2004; Ziebuhr 1999). But why this site is preferred is not fully explained — it's a documented pattern, not an established mechanism. The film shows what we've observed; "hot spot" labels a question that is still open.

? Throughout ON the chain looks exactly the same length even though polymer is being made continuously. Where do new units enter, and where do old ones leave?

Each peristaltic beat pops a fresh unit in at the growing end beside IcaA·D while the tip unit exits through IcaC toward the coat — production equals export, so net length stays constant. That's why the film reads as *flow* (a conveyor) rather than grow-then-shrink: if synthesis stopped you would see retraction; if export failed you would see a chain growing trapped inside.

? Catheter-associated S. aureus infections start with cells sticking to smooth plastic. Which chapter's state is compatible with sticking — and what would sequencing icaC from an implant strain that can't form biofilm show?

The ON / PIA/PNAG^{+} state — the deacetylated PNAG coat is exactly the adhesive layer on abiotic surfaces, so it's the coated cells that colonise devices. From a non-biofilm-forming isolate you'd expect IS256 inserted in *icaC* (a ~1.3 kb PCR product across the site), with function restorable by precise excision after passage — the same reversible switch the film animates.

Simplified for teaching — species-specific facts verified against primary literature (see README).

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